

Secure Centralized EV Charging Payment Gateway

Project Overview: This project implements a secure, centralized payment gateway for Electric Vehicle (EV) charging stations. It addresses security challenges in decentralized physical infrastructure by combining Lightweight Cryptography (LWC), Quantum Cryptographic Simulations, and Blockchain technology.

The system ensures:

- **Identity Security:** User and Franchise identities are protected using SHA-3 hashing.
- **Session Integrity:** Charging kiosks use an ASCON-inspired dynamic hashing (VFID) that rotates every 30 seconds to prevent session hijacking.
- **Quantum Risk Assessment:** A simulation of Shor's Algorithm demonstrates the vulnerability of classical RSA-based key exchanges.
- **Immutability:** All energy transactions are recorded on a mock Blockchain Ledger for transparent billing and dispute resolution.

Core Features:

- **SHA-3 Registration:** Users and Franchises generate unique 16-digit cryptographic identities.
- **Dynamic QR Scanning:** The EV owner interface includes a dropdown to select charging stations, simulating a QR scan by fetching the current LWC dynamic hash.
- **Grid Authority Audit:** A centralized terminal to view the user directory, franchise earnings, and the immutable transaction blocks.
- **Shor's Algorithm Simulation:** A factoring tool that can break large

composite numbers (like 97×83) to showcase the need for post-quantum crypto.

Tech Stack

- Backend: Python (Flask)
- Frontend: HTML5, CSS3 (Bootstrap 5), JavaScript
- Cryptography: hashlib (SHA-3, BLAKE2b), math (Shor's simulation)

Setup & Installation

4.1 Prerequisites

- Python 3.x installed on your system.
- Flask library.

4.2 Steps to Run

1. Extract the project files (BITSF463 Team 06.zip).
2. Install dependencies: `pip install flask`
3. Run the application: `python app.py`
4. Access the Gateway: Open your browser and navigate to <http://127.0.0.1:5000>

Assumptions & Limitations

- The blockchain is an in-memory simulation for demonstration and resets on server restart.
- Shor's algorithm is a hybrid simulation intended to demonstrate mathematical vulnerability rather than executing on a physical QPU. We have assumed 8051 for demonstration purposes.